Understanding APIs & RSS Feeds

A practical beginner's guide to the technology behind DeRexi Pulse

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THE BIG IDEA

APIs help one program ask another for information. RSS feeds publish new headlines in a consistent format. DeRexi Pulse uses both to help policy owners notice regulatory developments.

01 Why learn this?

DeRexi is a policy-governance assistant designed for financial institutions. Its Pulse feature introduces a regulatory-intelligence feed: one place where a policy owner can review newly published notices and guidance from public sources rather than repeatedly checking agency websites.

By the end, you should be able to define API and RSS, recognize JSON and XML, follow a data item from source to Pulse card, and explain why a human must confirm any compliance impact.

02 The two tools at a glance

	API	RSS FEED
Plain-English role	A way software requests information or actions	A standardized list of published updates
Common data format	JSON (also XML and others)	XML
Typical interaction	A client sends a request; server returns data	A feed reader periodically downloads the feed
Best use	Targeted access, search, filters, structured datasets	Headlines, announcements, publication dates
In Pulse	Read public structured government data	Collect newly issued public notices

EASY ANALOGY

API = ask the librarian for the exact book or record. RSS = read the library's continually updated bulletin of new arrivals.

03 How an API works

API stands for Application Programming Interface. An HTTP API exposes an address called an endpoint. A program requests it (often using GET) and receives an HTTP status plus a response body. A common body format is JSON: named fields and values.

Example only — illustrative JSON, not a real government response:

{ "title": "Draft guidance published", "published": "2026-11-05", "url": "https://example.org/notice" }

A status of 200 means the HTTP request was successful; 404 may mean the address was not found; 429 usually signals too many requests. None of these statuses establish that the content is a legally binding requirement.

04 How an RSS feed works

RSS stands for Really Simple Syndication. An RSS feed is an XML document with a channel and a list of items. Each item generally offers a title, a link, and publication metadata. A feed reader checks the feed repeatedly to find newly published items.

Illustrative RSS item: <item><title>New public notice</title><link>https://example.org/notice</link></item>

RSS is not an alert that magically arrives in DeRexi. The application must retrieve the feed or use a service that checks it, then decide which entries are new.

05 How the technology supports Pulse

SOURCE → FETCH → PARSE → NORMALIZE → VALIDATE → DISPLAY → REVIEW

Source — Public FDIC releases, Federal Reserve RSS, and NIST CSRC draft listings were explored for the project.

Fetch + parse — Retrieve a URL with HTTP; read a JSON response or parse XML feed items.

Normalize — Map differences into a common record: title, publisher, published date, type, original URL, summary.

Validate — Reject malformed data, handle missing fields, sanitize text, and retain primary-source links.

Display — Show clean Pulse cards with attribution and category filters; indicate unavailable sources.

Human review — An authorized owner reads the primary source and determines applicability, effective dates, and follow-up.

IMPORTANT DISTINCTION

A regulatory announcement, draft, guidance, and final rule are different kinds of publications. Pulse helps people notice them; it should not automatically label every new headline as a new binding policy obligation.

06 Our source strategy

PUBLIC SOURCE	WHAT WE LOOK FOR	CAUTION
FDIC	Banking announcements / press information	A webpage is not automatically a JSON API
Federal Reserve	RSS updates on regulatory and banking releases	Check feed currency and original link
NIST CSRC	Structured listings of draft material	Drafts are not final requirements

All integrations must use their current, officially published endpoints. External sources can change without notice. Free access does not guarantee unlimited requests or continued availability.

07 A five-minute hands-on exercise

1. Open the official Federal Reserve RSS feeds directory: https://www.federalreserve.gov/feeds/ . Pick a current feed URL.

2. View its XML. Find an item title, link, and publication field. Ask: what would show on a Pulse card?

3. Find a public JSON listing from an official documentation page. Look for keys (field names) and values.

4. Compare both formats: different syntax, but they can describe a similar publication.

5. Return to Pulse: identify where the source name, publication date, category, and link appear in the UI.

08 Quick comprehension check

Q: What is the difference between API and RSS? A: API is a software interaction interface; RSS is a published XML feed format.

Q: What is JSON? A: A structured format that represents data as values, objects and arrays.

Q: Can RSS be retrieved via HTTP? A: Yes; that is a common way applications read it.

Q: Does a new notice mean DeRexi should rewrite a policy? A: No; a policy owner first checks applicability and status.

09 Integration learning plan

Modeled on the instructor's PLAN.md: three small, testable tasks with success criteria and proof methods.

TASK	SUCCESS CRITERION	PROOF TO CAPTURE
1. Retrieve	Parse at least one public item with title and link	Dated response, source URL, terminal capture
2. Present	Display normalized Pulse items without crashing on missing fields	UI screenshot and test command/output
3. Verify + teach	Handle a failed feed and explain the flow to a new learner	Error-state proof, mini-tutorial, peer note

10 What has been done vs. what still needs proof

Exploration and development discussed: FDIC page, Federal Reserve RSS, NIST CSRC structured data, Pulse navigation and feed UI, and a card-based presentation direction. Earlier project notes mention an automated test run with 54 passes and 1 failure. This guide does not represent that historical discussion as a newly reproduced test result.

Still to verify: exact current source endpoint URLs, final filters and cards, captured working code, test cases and outputs, and a user walkthrough. Attach screenshots and commit links in learning/README.md after checking the working branch.

11 Good implementation habits

Use HTTPS and approved official hosts; set network timeouts; rate-limit requests; retry carefully; cache results; escape remote content; preserve source links and timestamps; log failures without exposing credentials. Treat downloaded content as untrusted text, not commands or policy instructions.

12 Reflection

The biggest learning shift is recognizing that collecting information and making a compliance decision are separate processes. APIs and RSS solve the discovery problem. Pulse organizes information and helps route attention. Human policy owners evaluate significance and determine whether any policy change is actually necessary.

Further reading

Instructor templates: github.com/kakradetome/dae-6-month-cohort-2/tree/main/learning

HTTP / JSON: developer.mozilla.org/en-US/docs/Web/HTTP/Overview • developer.mozilla.org/en-US/docs/Learn_web_development/Core/Scripting/JSON

RSS: rssboard.org/rss-specification • Federal Reserve: federalreserve.gov/feeds/ • NIST: csrc.nist.gov/ • FDIC: fdic.gov/news